AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #3

Activity Name: Seven Segment Display Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**28|07|2026**
Submission Date:	**28|07|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to interface Seven Segment Display. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
#include <LPC21xx.h>

// Delay Function
void delay(unsigned int t)
{
    unsigned int i, j;
    for(i = 0; i < t; i++)
        for(j = 0; j < 6000; j++);
}

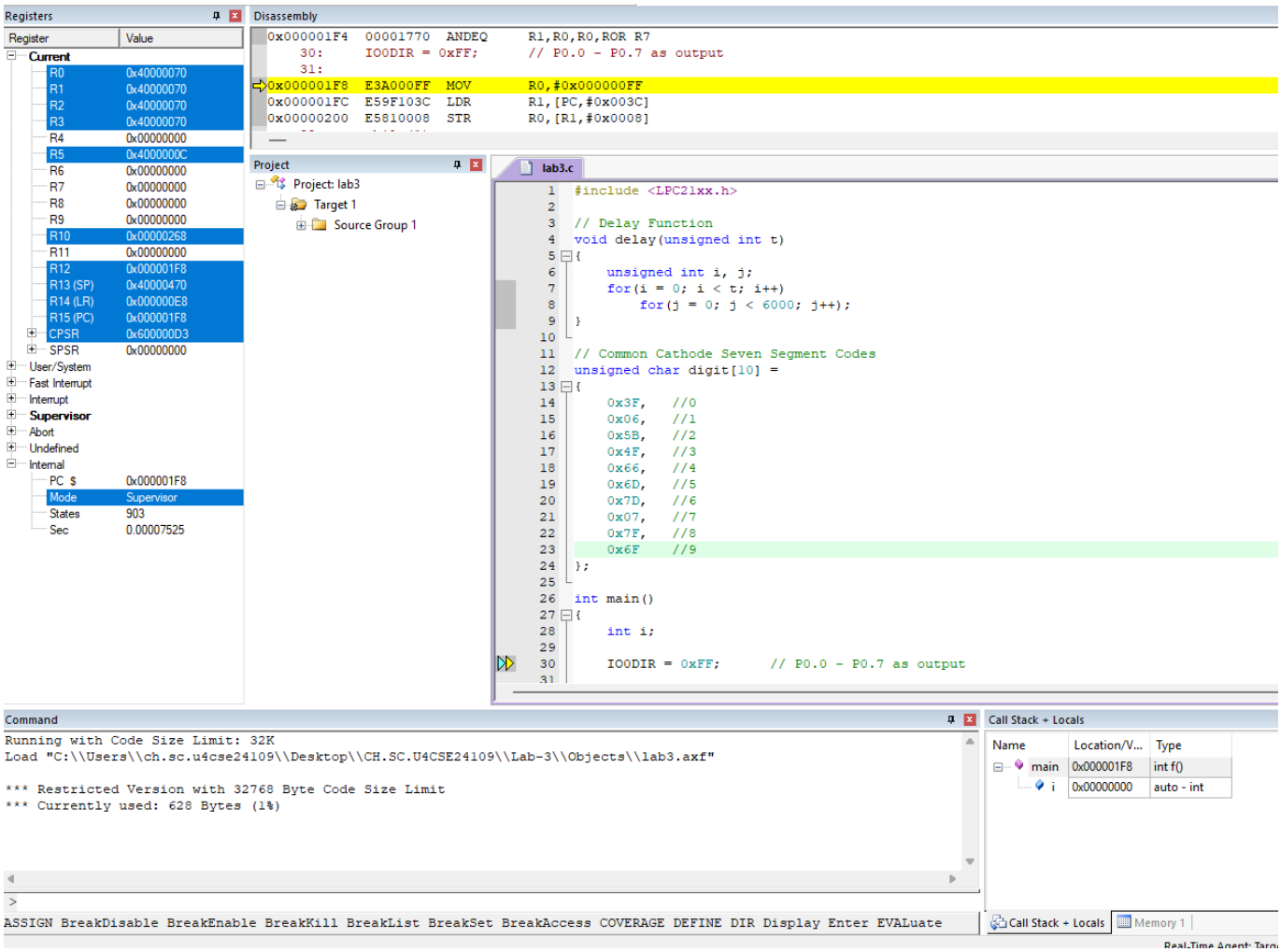
// Common Cathode Seven Segment Codes
unsigned char digit[10] =
{
    0x3F,   //0
    0x06,   //1
    0x5B,   //2
    0x4F,   //3
    0x66,   //4
    0x6D,   //5
    0x7D,   //6
    0x07,   //7
    0x7F,   //8
    0x6F    //9
};

int main()
{
    int i;

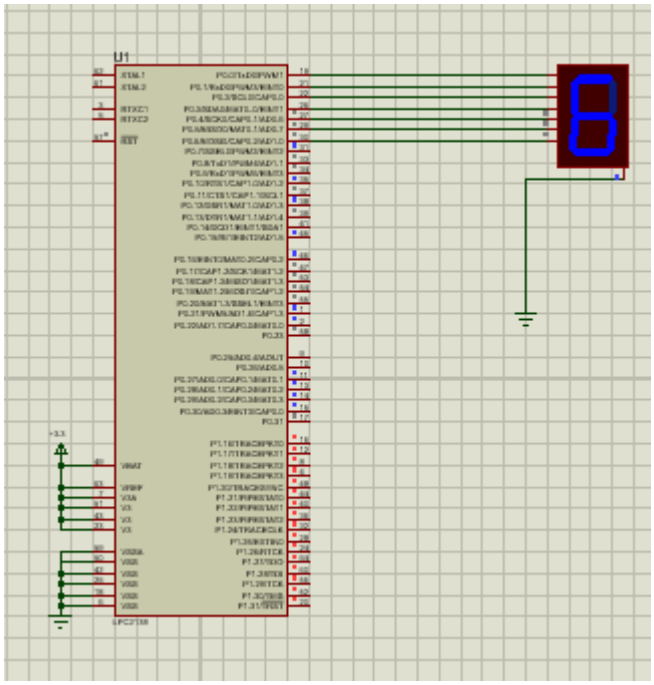
    IO0DIR = 0xFF;      // P0.0 - P0.7 as output

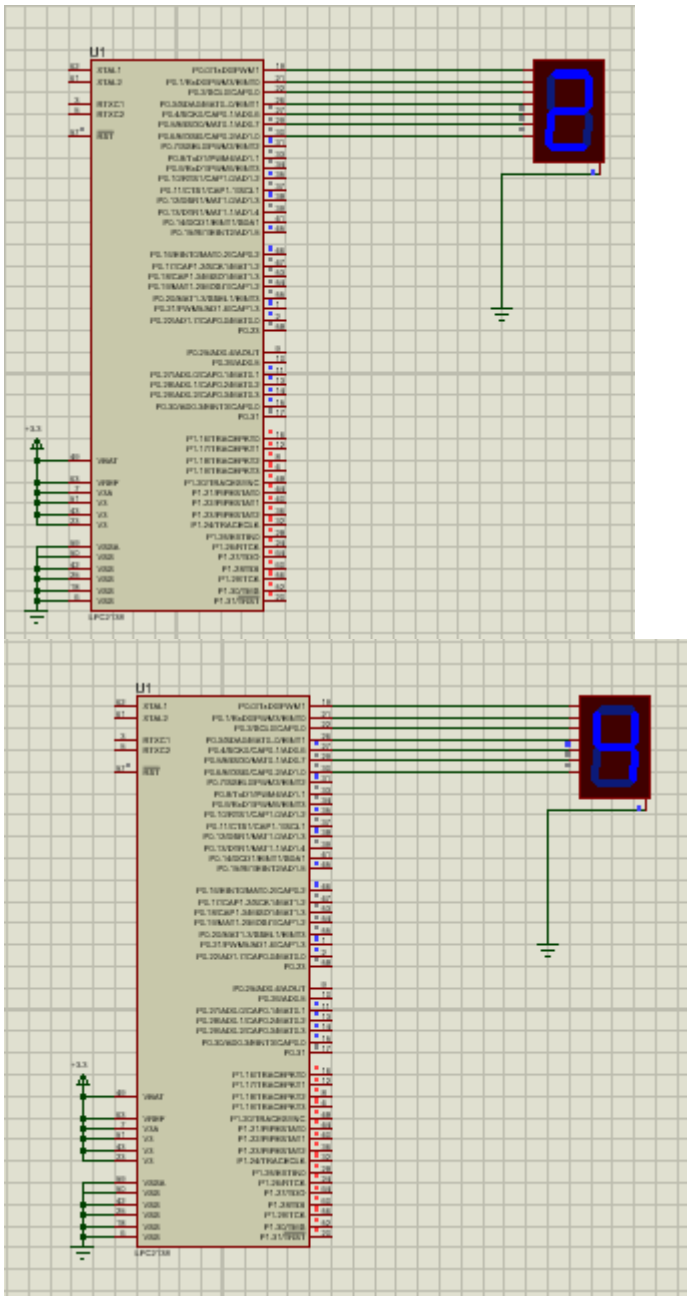
    while(1)
    {
        for(i = 0; i < 10; i++)
        {
            IO0CLR = 0xFF;          // Clear previous value
            IO0SET = digit[i];      // Display current digit
            delay(10);
        }
    }
}
```

OUTPUT WINDOW:


Registers
Register
Value
Current
R0 0x40000070
R1 0x40000070
R2 0x40000070
R3 0x40000070
R4 0x00000000
R5 0x4000000C
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x00000268
R11 0x00000000
R12 0x000001F8
R13 (SP) 0x40000470
R14 (LR) 0x000000E8
R15 (PC) 0x000001F8
CPSR 0x600000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x000001F8
Mode Supervisor
States 903
Sec 0.00007525
Disassembly
0x000001F4 00001770 ANDEQ R1,R0,R0,ROR R7
30: IO0DIR = 0xFF; // P0.0 - P0.7 as output
31:
0x000001F8 E3A000FF MOV R0,#0x000000FF
0x000001FC E59F103C LDR R1,[PC,#0x003C]
0x00000200 E5810008 STR R0,[R1,#0x0008]
Project
Project: lab3
Target 1
Source Group 1
lab3.c
1 #include <LPC21xx.h>
2
3 // Delay Function
4 void delay(unsigned int t)
5 {
6 unsigned int i, j;
7 for(i = 0; i < t; i++)
8 for(j = 0; j < 6000; j++);
9 }
10
11 // Common Cathode Seven Segment Codes
12 unsigned char digit[10] =
13 {
14 0x3F, //0
15 0x06, //1
16 0x5B, //2
17 0x4F, //3
18 0x66, //4
19 0x6D, //5
20 0x7D, //6
21 0x07, //7
22 0x7F, //8
23 0x6F //9
24 };
25
26 int main()
27 {
28 int i;
29
30 IO0DIR = 0xFF; // P0.0 - P0.7 as output
31
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ch.sc.u4cse24109\\Desktop\\CH.SC.U4CSE24109\\Lab-3\\Objects\\lab3.axf"
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 628 Bytes (1%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Locals
Name Location/V... Type
main 0x000001F8 int f()
i 0x00000000 auto - int
Call Stack + Locals
Memory 1

OUTPUT:


U1
LPC2138


U1
+3.3
LPC2138

U1
+3.3
LPC2138